Appendix C: Flagship Proofs for ORIUS

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Scope Note. This standalone article is the reviewer-facing integrated proof packet for ORIUS and the registry-canonical flagship-core execution draft. It presents the battery witness ladder T1–T8 first, then the defended ORIUS-wide extension surfaces T11 and T_trajectory_PAC. The active theorem registry remains authoritative for defense tiers: T3 is presented through the flagship T3a theorem plus the supporting T3b corollary, T5 remains definition-level within the ladder, and T8 remains a supporting sequence-level result even though it is included here for reviewer continuity. The full integrated proof source remains `appendices/app_c_full_proofs.tex`.

C.1 Strict Topological Notation and Dynamics

Every proof below uses the explicit Assumption Register in Appendix B together with the active flagship theorem surfaces only.

Symbol	Type	Meaning
x_t	$\mathbb{R}$	Latent true state at step t.
$\hat{x}_t$	$\mathbb{R}$	Observed state at step t.
U_t	Set	Observation-consistent / reliability-adjusted uncertainty set at step t.
$\mathcal{A}_t$	Set	Tightened battery safe-action set used by the DC3S shield.
$\mathcal{A}_t^{\text{safe}}(U_t)$	Set	Typed safe-action set sound for the uncertainty set U_t.
$\Delta(a_t)$	$\mathbb{R}$	One-step state increment induced by action a_t.
ϵ_t	$\mathbb{R}$	One-step model residual with $\lvert\epsilon_t\rvert \leq \epsilon_{\text{model}}$ under A1.
$\mathcal{S}$	Set	Generic domain safe set.
$[s_{\min}, s_{\max}]$	Interval	Battery safe interval.
$\mathcal{H}_t$	σ-algebra	Causal observation history up to step t.
Z_t	$\{0,1\}$	True-state violation indicator at step t.
V	$\mathbb{N}$	Total true-state violation count across the horizon.
w_t	$[0,1]$	Reliability score at step t.
$\bar{w}$	$[0,1]$	Episode-average reliability $T^{-1}\sum_{t=1}^{T} w_t$.
$\delta_{\text{bnd},t}$	$\mathbb{R}_+$	Margin from the issued certificate interval to the nearest SOC boundary.
τ_t	$\mathbb{N}$	Certifiable battery validity horizon or lower-bound horizon.
H	$\mathbb{N}$	Trajectory horizon for the PAC certificate.
n_{cal}	$\mathbb{N}$	Calibration sample size.
n_{eff}	$\mathbb{N}$	Effective reliability-deflated calibration size.
ε_{fs}	$\mathbb{R}_+$	Finite-sample conformal correction term.
$H_{\max}$	$\mathbb{N}$	Maximum certifiable safe horizon under the trajectory certificate.

C.2 T1 — OASG Existence

Theorem 1 (T1: OASG Existence)**.** ***Assumptions.*** *A1, A2, A4, A11, A12.*
Statement. *For any dropout rate $d > 0$ and any optimal arbitrage controller, there exists a time step t^* such that*

$$\phi_{\text{obs}}(a_{t^*}) = 1 \qquad \textit{and} \qquad \phi_{\text{true}}(a_{t^*}) = 0.$$

That is, the released action is safe in observation but unsafe in truth.

Proof. By A11, the battery witness controller can be driven to a boundary-sensitive state under admissible demand, so there exists a step where the true SOC lies within a boundary distance δ_{bnd} of the constraint surface. By A2 and the standing dropout rate $d > 0$, degraded telemetry admits a non-trivial observation gap on an admissible fault window. A12 keeps the fault process conditionally independent of the controller action stream given the history, so the boundary-sensitive witness step and the admissible telemetry fault can co-occur on the same realized history.

Choose such a step t^*. The controller releases the action using the observed state and therefore treats that action as feasible on the observed side of the boundary. The true state, however, is displaced by the admissible observation gap beyond the boundary-facing side of the safe set, so the same released action crosses the true constraint surface. A4 supplies the shared one-step dynamics

interpretation needed to compare the observed-state and true-state outcomes for that released action. Hence

$$\phi_{\text{obs}}(a_{t^*}) = 1 \qquad \text{and} \qquad \phi_{\text{true}}(a_{t^*}) = 0,$$

which is exactly the battery witness-row OASG phenomenon. ■

C.3 T2 — Safety Preservation

Theorem 2 (T2: Safety Preservation)**.** ***Assumptions.*** *A1, A2, A3, A4, A5, A7.*
Statement. *If the DC3S shield selects $a_t \in \mathcal{A}_t$ and the current true state satisfies $x_t \in U_t$, then the released action is one-step safe:*

$$x_t \in U_t \;\; \textit{and} \;\; a_t \in \mathcal{A}_t \quad \Longrightarrow \quad x_{t+1} \in [s_{\min}, s_{\max}].$$

No episode-level expectation claim is made in this theorem.

Proof. Fix a step t and assume the theorem hypotheses. Because $x_t \in U_t$, the current true state lies in the observation-consistent battery tube, so

$$|x_t - \hat{x}_t| \le m_t.$$

Because $a_t \in \mathcal{A}_t$, the shielded observed-state prediction is required to stay inside the absorbed tightened interval:

$$s_{\min} + m_t + \epsilon_{\text{model}} \le \hat{x}_t + \Delta(a_t) \le s_{\max} - m_t - \epsilon_{\text{model}}.$$

Now write the true successor using A4:

$$x_{t+1} = x_t + \Delta(a_t) + \epsilon_t = \hat{x}_t + \Delta(a_t) + (x_t - \hat{x}_t) + \epsilon_t.$$

Using A1 and the tube bound gives the lower estimate

$$x_{t+1} \ge (\hat{x}_t + \Delta(a_t)) - |x_t - \hat{x}_t| - |\epsilon_t| \ge (s_{\min} + m_t + \epsilon_{\text{model}}) - m_t - \epsilon_{\text{model}} = s_{\min}.$$

The same substitution yields the upper estimate

$$x_{t+1} \le (\hat{x}_t + \Delta(a_t)) + |x_t - \hat{x}_t| + |\epsilon_t| \le (s_{\max} - m_t - \epsilon_{\text{model}}) + m_t + \epsilon_{\text{model}} = s_{\max}.$$

Therefore

$$s_{\min} \le x_{t+1} \le s_{\max},$$

so the released action is one-step safe. The proof uses only current-step information and therefore preserves the causal certificate discipline required by A7. ■

C.4 T3a — ORIUS Core Envelope Derivation

Lemma 0.1 (Aggregation under a predictable risk budget)**.** *If a predictable nonnegative sequence r_t satisfies*

$$\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le r_t \qquad \textit{for all } t,$$

then

$$\mathbb{E}[V] \le \sum_{t=1}^{T} r_t.$$

Proof. Because Z_t is an indicator,

$$\mathbb{E}[Z_t \mid \mathcal{H}_t] = \mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \leq r_t.$$

Taking unconditional expectations preserves the inequality, and summing across the horizon yields $\mathbb{E}[V] \leq \sum_{t=1}^{T} r_t$. No temporal independence assumption is required. ■

Theorem 3 (ORIUS Core Bound)**.** ***Assumptions.*** *A1, A2, A4, A5, A6, A7, A9.*
Statement. *Under the battery witness-row predictable one-step risk budget*

$$\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \leq \alpha(1 - w_t),$$

the expected total violation count under the DC3S shield satisfies

$$\mathbb{E}[V] \leq \alpha(1 - \bar{w})T,$$

where $\bar{w} = T^{-1} \sum_{t=1}^{T} w_t$ *is the episode-average reliability.*

Proof. Apply Lemma 0.1 with $r_t = \alpha(1 - w_t)$. Then

$$\begin{aligned}
\mathbb{E}[V] &\leq \sum_{t=1}^{T} \alpha(1 - w_t) \\
&= \alpha \sum_{t=1}^{T} (1 - w_t) \\
&= \alpha T \left(1 - \frac{1}{T} \sum_{t=1}^{T} w_t\right) \\
&= \alpha(1 - \bar{w})T.
\end{aligned}$$

Thus the flagship T3a surface is an aggregation argument over an explicit predictable risk-envelope contract rather than a standalone corollary of marginal conformal coverage. ■

Registry Note. The active theorem registry treats the theorem above as flagship-defended $T3a$. The supporting corollary $T3b$ remains in `appendices/app_c_full_proofs.tex` and is not reproduced in this flagship-only appendix.

C.5 T4 — No Free Safety

Theorem 4 (T4: No Free Safety)**.** ***Assumptions.*** *A1, A2, A4, A11.*
Statement. *For any fixed-margin quality-ignorant controller* π *and any dropout rate* $d > 0$*, there exists an admissible fault sequence and a time* $t^\star$ *such that*

$$\phi_{\text{obs}}(\pi(\hat{x}_{t^\star})) = 1 \qquad \textit{and} \qquad \phi_{\text{true}}(\pi(\hat{x}_{t^\star})) = 0.$$

Equivalently, the controller incurs at least one true-state violation on the constructed admissible sequence.

Proof. Fix a quality-ignorant controller π. By definition, its effective safety margin does not depend on the reliability score, so there exists a constant $m^\pi \geq 0$ that upper-bounds the observation buffer implicitly used by the controller. By A11, the battery witness controller can be driven through a boundary-sensitive baiting phase in which the state approaches the constraint surface while telemetry remains nominal.

Because $d > 0$, A2 admits an admissible degraded-observation fault window with a non-trivial observation gap. Choose a step $t^\star$ in that window where the accumulated observation error exceeds the controller's fixed margin plus the boundary distance available on the baited state. At that step the observed state still appears feasible relative to the controller's fixed margin, but the true state has shifted across the boundary-facing side of the admissible set by more than the controller can compensate. A4 again supplies the shared one-step battery dynamics interpretation for the released action.

Therefore the action $\pi(\hat{x}_{t^\star})$ remains observed-state legal while creating a true-state violation. This yields an observation-action safety gap and hence at least one true-state violation on the constructed admissible sequence. The theorem is intentionally class-scoped to fixed-margin, quality-ignorant controllers on the battery witness row. ■

C.6 T5 — Certificate Validity Horizon

Definition 5 (T5: Certificate Validity Horizon)**.** ***Assumptions.*** *A4, A7, A9.*
Statement. *The battery validity horizon τ_t is the largest integer horizon whose forward tube remains inside the SOC constraint interval. This row is constructive and definition-level inside the integrated ladder.*

Proof. Fix the issued battery interval and the certified safe action. The forward tube at horizon Δ is obtained by propagating that action under the battery one-step dynamics while adding the worst-case sub-Gaussian radius used by the defended runtime. The set of admissible horizons is therefore exactly the set of integers for which the resulting tube stays inside $[s_{\min}, s_{\max}]$. The maximal such integer is τ_t, and the executable helper `certificate_validity_horizon()` computes precisely that constructive object. ■

C.7 T6 — Certificate Expiration Bound

Proposition 6 (T6: Certificate Expiration Bound)**.** ***Assumptions.*** *A4, A7, A9, and confidence level $\delta \in (0, 1)$.*
Statement. *The battery certificate horizon satisfies the lower bound*

$$\tau_t \geq \left\lfloor \frac{\delta_{\mathrm{bnd},t}^2}{2\sigma_d^2 \log(2/\delta)} \right\rfloor .$$

Proof. For a σ_d-sub-Gaussian disturbance, the reflection-principle / first-passage tail bound gives

$$\mathbb{P}(|S_\Delta| \geq \delta_{\mathrm{bnd},t}) \leq 2 \exp\left(-\frac{\delta_{\mathrm{bnd},t}^2}{2\sigma_d^2 \Delta} \right) .$$

Demanding the right-hand side be at most δ yields

$$\Delta \leq \frac{\delta_{\mathrm{bnd},t}^2}{2\sigma_d^2 \log(2/\delta)} .$$

Because the validity horizon is integer-valued, flooring gives the claimed lower bound. The theorem-facing helper `certificate_expiration_bound()` implements this exact closed form and records its explicit side conditions in the returned theorem contract. ■

C.8 T7 — Feasible Fallback Existence

Proposition 7 (T7: Feasible Fallback Existence)**.** ***Assumptions.*** *A1, A4, A8.*
Statement. *For any safe battery state, the defended fallback surface is piecewise:*

either zero dispatch safely holds on the interior,

or a boundary-aware safe-landing action safely moves the SOC inward.

If neither case can be certified under the active constraints, the runtime fails closed rather than claiming a false fallback theorem.

Proof. If the current SOC has enough interior slack, zero dispatch gives $x_{t+1} = x_t + \epsilon_t$ with $|\epsilon_t| \leq \epsilon_{\text{model}}$, so the next state remains inside the admissible interval. This is the hold case.

If the current SOC is safe but too close to a boundary for zero dispatch to be certified, the safe-landing controller computes a charge or discharge action that moves the SOC toward the interior safe zone while respecting the active battery power, ramp, and efficiency constraints. The theorem-facing validator checks the corresponding successor interval under the bounded model error and accepts this case only when the entire interval remains safe.

If neither the hold case nor the safe-landing case is certifiable, the validator returns infeasible. The defended theorem therefore proves the piecewise hold-or-safe-landing battery surface and does not claim a universal fallback guarantee beyond those executable cases. ■

C.9 T8 — Graceful Degradation Dominance

Proposition 8 (T8: Graceful Degradation Dominance)**.** ***Assumptions.*** *A1–A5, A8.*
Statement. *For graceful and uncontrolled violation sequences produced under the same admissible fault trace,*

$$z_t^{\text{graceful}} \leq z_t^{\text{uncontrolled}} \ \forall t \quad \Longrightarrow \quad V^{\text{graceful}} \leq V^{\text{uncontrolled}}.$$

Strict improvement is claimed only when the graceful sequence is strictly smaller on at least one step.

Proof. Summing the stepwise dominance inequality over the shared admissible fault trace gives

$$\sum_{t=1}^{T} z_t^{\text{graceful}} \leq \sum_{t=1}^{T} z_t^{\text{uncontrolled}},$$

which is exactly the claimed cumulative-count dominance statement. Strict inequality follows if the graceful sequence is strictly smaller on at least one step. This is the supporting T8 surface in the integrated ladder: a paired-sequence comparison and nothing stronger. ■

C.10 T11 — Typed Structural Transfer

Local T11 Notation. In this section, U_t denotes the constructed observation-consistent state set and $\mathcal{A}_t^{\text{safe}}(U_t)$ denotes the tightened safe-action set that is sound for that uncertainty set.

Theorem 9 (Typed structural transfer theorem, T11)**.** ***Assumptions.*** *For every step t:*

1. *the constructed observation-consistent state set satisfies $\mathbb{P}[x_t \in U_t \mid \mathcal{H}_t] \geq 1 - \alpha$;*
2. *the tightened safe-action set is sound for that uncertainty set, in the sense that every action in $\mathcal{A}_t^{\text{safe}}(U_t)$ keeps the next state inside $\mathcal{S}$ for all $x \in U_t$ and all admissible disturbance;*
3. *the repair map returns an element of $\mathcal{A}_t^{\text{safe}}(U_t)$ whenever that set is non-empty;*
4. *when the tightened safe-action set is empty, the fallback action is admissible and preserves safety from any current safe state.*

Statement. *Let $\mathcal{D}$ be a domain instantiation for the degraded-observation safe-control problem. Then the repaired action produced by the ORIUS kernel satisfies the same one-step safety-preservation statement as the reference theorem:*

$$\mathbb{P}[x_{t+1} \in \mathcal{S} \mid \mathcal{H}_t] \geq 1 - \alpha.$$

No broader episode-level claim is inherited unless the new domain separately supplies an explicit predictable per-step risk budget.

Proof. Fix a step t and condition on the observation history $\mathcal{H}_t$. On the event

$$E_t = \{x_t \in U_t\},$$

Assumption (2) implies that every action in $\mathcal{A}_t^{\text{safe}}(U_t)$ preserves membership in the safe set at the next step. If the tightened safe-action set is non-empty, then Assumption (3) ensures that the repaired action returned by the kernel lies in that set and hence preserves safety on E_t. If the tightened safe-action set is empty, then Assumption (4) ensures that the fallback action preserves safety from any current safe state. Therefore,

$$\mathbb{P}[x_{t+1} \in \mathcal{S} \mid \mathcal{H}_t] \geq \mathbb{P}[E_t \mid \mathcal{H}_t] \geq 1 - \alpha,$$

which is exactly the forward one-step transfer claim. The converse remains a separate structural failure proposition and is not part of active T11. ■

C.11 T_trajectory_PAC — PAC Trajectory Certificate

Theorem 10 (T_trajectory_PAC: PAC Trajectory Certificate)**.** ***Assumptions.*** *A1, A4, A5, A9, and Bonferroni/union-bound aggregation over the horizon. For horizon H, calibration size n_{cal}, and confidence $1 - \delta$:*

$$\mathbb{P}\big[\textit{all } H \textit{ steps safe}\big] \geq 1 - H\alpha(1 - \bar{w}) - \varepsilon_{\text{fs}} - \delta/2,$$

where

$$\varepsilon_{\text{fs}} = \sqrt{\ln(4/\delta)/(2n_{\text{eff}})} \qquad \textit{and} \qquad n_{\text{eff}} = n_{\text{cal}} \cdot \min_t w_t.$$

The maximum certifiable horizon is

$$H_{\max} = \left\lfloor \frac{1 - \varepsilon_{\text{fs}} - \delta/2}{\alpha(1 - \bar{w})} \right\rfloor.$$

Proof. **Step 1 (per-step risk).** By T3a,

$$\mathbb{P}[Z_t = 1 \mid \mathcal{H}_t] \le \alpha(1 - w_t).$$

Step 2 (trajectory aggregation by union bound). Let $E_t = \{Z_t = 1\}$ denote a stepwise true-state violation. Then

$$\mathbb{P}\Big[\bigcup_{t=1}^{H} E_t\Big] \le \sum_{t=1}^{H} \mathbb{P}[E_t] \le \sum_{t=1}^{H} \alpha(1 - w_t) \le H\alpha(1 - \bar{w}).$$

Step 3 (finite-sample correction). The conformal quantile $\hat{q}$ estimated from n_{cal} points satisfies

$$\mathbb{P}[\hat{q} < q_{\text{true}}] \le \varepsilon_{\text{fs}}.$$

We deflate the effective calibration size by $\min_t w_t$ to account for the worst-case reliability in the trajectory.

Step 4 (combining).

$$\mathbb{P}[\exists\, t \in [H] : \text{violation}] \le \underbrace{H\alpha(1 - \bar{w})}_{\text{risk budget}} + \underbrace{\varepsilon_{\text{fs}}}_{\text{calibration error}} + \underbrace{\delta/2}_{\text{exit-time budget}}.$$

Hence

$$\mathbb{P}[\text{all safe}] \ge 1 - H\alpha(1 - \bar{w}) - \varepsilon_{\text{fs}} - \delta/2.$$

Setting the right-hand side nonnegative and solving for H gives the stated $H_{\max}$ formula. This is the defended Bonferroni/union-bound certificate and nothing stronger; no martingale or Ville-style strengthening is claimed here. ■